Name

Mathematics Stage 5 Core

Significant Figures and Scientific Notation

Book 1

Version: 261006

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-MAG-C-01 solves measurement problems by using scientific notation to represent numbers and rounding to a given number of significant figures

Identify and describe very small and very large measurements

- Identify and describe the meaning of common prefixes, such as milli, centi and kilo
- Establish the meaning of prefixes for very small or very large measurement units

Estimate and round numbers to a specified degree of accuracy

- Apply the language of estimation appropriately, including the terms *rounding*, *approximate* and *level of accuracy*
- Round numbers to a specified number of significant figures
- Examine the effect that truncating or rounding during calculations has on the accuracy of the results

Express numbers in scientific notation

- Recognise the need for notation to express very large or very small numbers
- Represent numbers using scientific notation in practical contexts
- Order numbers expressed in scientific notation
- Represent numbers expressed in scientific notation as a decimal
- Estimate the value of calculations involving scientific notation by applying knowledge of index laws
- Solve problems with calculations involving scientific notation using digital tools

The Metric System

ⓘ Metric Prefixes

- The **metric** system of measurement bases everything on **powers of 10**.
- The **prefix** of a unit of measurement tells you how large or small it is.

Prefix		Meaning	Index notation	Number	Fraction
Giga	G	Billion	10^9	$1\,000\,000\,000$	-
Mega	M	Million	10^6	$1\,000\,000$	-
kilo	k	Thousand	10^3	1000	-
(no prefix)		One	1	1	1
centi	c	Hundredth	10^{-2}	0.01	$\frac{1}{100}$
milli	m	Thousandth	10^{-3}	0.001	$\frac{1}{1000}$
micro	μ	Millionth	10^{-6}	$0.000\,001$	$\frac{1}{1\,000\,000}$
nano	n	Billionth	10^{-9}	$0.000\,000\,001$	$\frac{1}{1\,000\,000\,000}$

- Unusual units, e.g. centilitres, are used in other countries and laboratory, pharmaceutical, or medical fields.

ⓘ SI Units

- **SI units** are internationally agreed upon units of measurement.
- The prefix of the SI unit tells you how large or small the unit of measurement is.

Quantity	SI unit
Length	Metre (m)
Capacity	Litre (L)
Mass	Gram (g)
Time	Second (s)
Energy	Joule (J)

- **Note**: for time, we use metric units only when using smaller units of measurement; millisecond is widely used; kilosecond is only used in specific scientific and computing fields.

1 Complete the table.

Prefix		Meaning	Index notation
Giga	G	Billion	10^9
Mega	M	Million	10^6
kilo	k	thousand	10^3
(no prefix)		One	1
centi	C	hundredth	10^{-2}
milli	m	thousandth	10^{-3}
micro	μ	millionth	10^{-6}
nano	n	Billionth	10^{-9}

2 Complete the table.

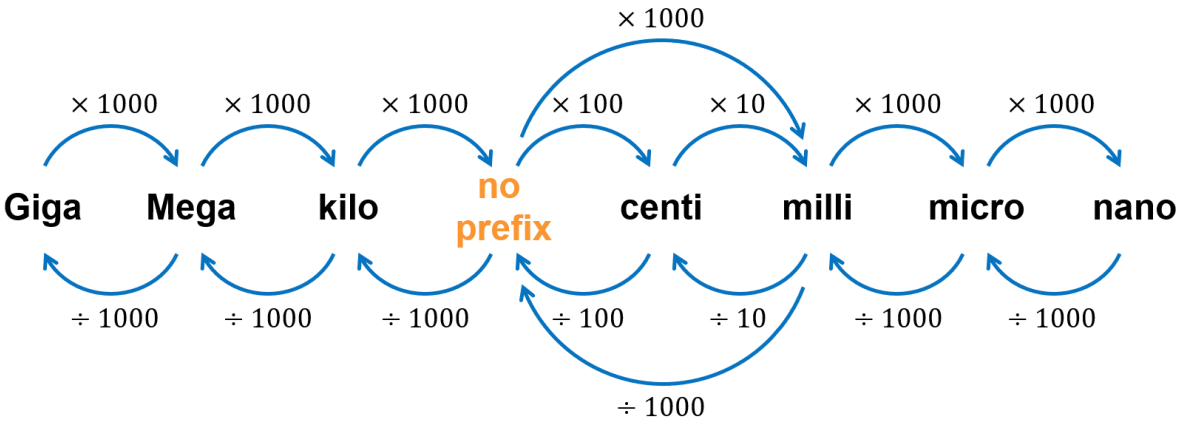
Prefix		Meaning	Index notation
kilo	k	Thousand	10^3
(no prefix)		One	1 or 10^0
nano	n	Billionth	10^{-9}
Giga	G	Billion	10^9
milli	m	Thousandth	10^{-3}
Micro	μ	Millionth	10^{-6}
centi	c	Hundredth	10^{-2}
Mega	M	Million	10^6

Metric Conversions

ⓘ Metric Conversions

If converting to a smaller unit multiply

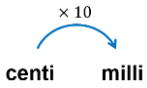
If converting to a larger unit, divide



- Note: a tonne (t) is 1000 kg or 1 Mg

Example Convert metric units

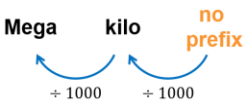
Convert 35 cm to mm



$35 \text{ cm} \times 10$

$= 350 \text{ mm}$

Convert 520 000 L to ML

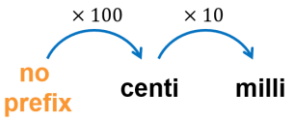


$520\,000 \text{ L} \div 1000 \div 1000$

$= 0.52 \text{ ML}$

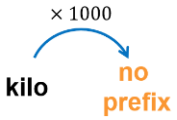
Guided Practice

a 25 m to mm



25 000 mm

b 1500 kg to g



1 500 000 g

c 6700 g to kg



6.7 kg

d 11.55 L to mL

11 550 mL

e 8 500 000 mL to kL

8.5 kL

f 80 mL to L

0.08 L

1 Convert the following.

a	2 L= ..... mL 2000	**b**	12 kL = ... L 12 000	**c**	9 kL = ... mL 9 000 000
d	7800 kL = ... L 7 800 000	**e**	50 L = ... mL 50 000	**f**	300 kL = ... mL 300 000 000
g	6100 L = ... kL 6.1	**h**	400 mL = ... L 0.4	**i**	210 000 mL = ... kL 0.21
j	3 t = ... kg 3000	**k**	45 kg = ... g 45 000	**l**	76 t = ... kg 76 000
m	8100 kg = ... g 8 100 000	**n**	4 t = ... g 4 000 000	**o**	0.52 t = ... kg 520
p	6800 g = ... kg 6.8	**q**	300 kg = ... t 0.3	**r**	2300 g = ... kg 2.3

2 Fill in the table.

	km	m	cm	mm
a	1	1000	100 000	1000 000
b	0.001	1	100	1000
c	0.000 01	0.01	1	10
d	0.000 001	0.001	0.1	1
e	0.000 01	0.01	1	10
f	0.000 17	0.17	17	170
g	0.001 7	1.7	170	1700
h	1.07	1070	107 000	1 070 000
i	0.07	70	7000	70 000
j	0.000 07	0.07	7	70
k	0.000 000 7	0.000 7	0.07	0.7
l	0.000 25	$\frac{1}{4}$	25	250

3 There are 20 litres of a chemical stored in a container.

a What amount of chemical remains if 750 mL is removed from the container? Answer in litres.

19.25 L

b How many containers are required to hold a kilolitre of the chemical?

50 containers

4 The length of the Murray River is 2575 km. The length of the Hawkesbury River is 80 000 m. What is the difference in their lengths? Answer in metres.

2 495 000 m

5 Christopher bought 3 kg of sultanas. What mass of sultanas remains if he ate 800 grams? Answer in kilograms.

2.2 kg

6 A truck is loaded with 3000 bricks, each of mass 4 kg. Find the total mass (in t) of the bricks.

12 t

7 The mass of a box of cereal is 375 g. What is the total mass (in kg) of 15 boxes of cereal?

5.625 kg

8 Find the total mass (in g) of five items of mass 250 mg,1100 mg, 0.7 g, 5.95 g, and 15.4 g.

23.4 g

9 An empty plastic box has mass 750 g. It is packed with 8 bottles of drink, each of mass 1.25 kg. Calculate the total mass (in g) of the packed box.

10 750 g

10 A cyclist travels to and from work over a 1200-metre long bridge. Calculate the distance travelled in a week if the cyclist works for 5 days. Answer in kilometres.

12 km

11 Convert the following.

a 7.2 Gg to t 7200 t	**b** 2.9 cg to ng 29 000 000 ng	**c** 9 kL to mL 9 000 000 mL
d 4.3 cm to μm 43 000 μm	**e** 5.3 ms to ns 5 300 000 ns	**f** 9 kg to mg 9 000 000 mL

12 Warragamba dam holds approximately 2 580 000 ML of water at full capacity.
How many gigalitres is this?

2580 GL

13 A computer can access its memory in 24 ns. Convert this to microseconds.

24 000 μs

14 Tera- (T) is the metric prefix for trillion ($10^{12} = 1\ 000\ 000\ 000\ 000$).
The distance from Mars to the Sun is 0.228 Tm. Convert this distance to kilometres.

228 000 000 km

15 The mass of a hydrogen atom is 1.67×10^{-24} g.
What is the mass in nanograms of 1 million hydrogen atoms?

1.67×10^{-9} ng

16 The total solar energy hitting Australia in one minute is estimated (by ChatGPT) to be 7.22×10^{19} Joules. Australia's annual energy consumption is 6500 Petajoules (10^{18}).
Assuming we can harness all the solar energy, for how long would we need to collect solar energy to power the whole country for a year?

$6500 \times 10^{18} \div (7.22 \times 10^{19})$

Around 90 minutes.

Significant Figures

ⓘ Significant Figures

- **Significant figures** are meaningful digits in a number.
- It's impossible to cut a piece of timber to a length of 1.333 333… m.
- The digits after a certain figure, or digit, are meaningless.

- **Rules for significant figures:**
  1. The first significant figure is the first non-zero digit (reading from left to right).
  2. All non-zero digits are significant.
  3. Zeros between non-zero digits are significant.
  4. Zeros on the end of a decimal are significant.
  5. Zeros on the end of a whole number *may* be significant.
     Treat them as significant until we learn scientific notation.



🔍 **Identify** number of significant figures

a	456	**i**	0.04	**q**	1.45006	**y**	4.00067
b	450	**j**	0.004	**r**	450067	**z**	0.00067
c	406	**k**	0.00456	**s**	45067	**aa**	0.0067
d	400	**l**	0.456	**t**	4506.7	**bb**	6.0007
e	40	**m**	0.406	**u**	450.67	**cc**	0.6007
f	4	**n**	0.450	**v**	45.067	**dd**	0.0607
g	0.4	**o**	0.4500	**w**	45.0067	**ee**	1.06070
h	0.40	**p**	0.45006	**x**	4.50067	**ff**	1.060700

Round to a Significant Figure

- To round to n significant figures:
  1. Identify the nth significant figure.
  2. Determine the place value of this digit.
  3. Round to this place value.

Example Round numbers to a specific number of significant figures

Round 56718 to one significant figure.

1. The first significant figure is digit 5,
2. representing ten thousand.
3. We round to the nearest ten thousand.

$56718 \approx 60000\ (1\text{ s.f.})$

Round 0.0068463 to two significant figures.

1. The 2nd significant figure is digit 8,
2. the 4th decimal place.
3. We round to the 4th decimal place.

$0.0068463 \approx 0.0068\ (2\text{ s.f.})$

Guided Practice

Round these numbers to the required significant figures

a 428.3 (1 s.f.)

1st significant figure: ……

Round to nearest ……

$428.3 \approx$ …… (1 s.f.)

400

b 629.51 (3 s.f.)

3rd significant figure: ……

Round to nearest ……

$629.51 \approx$ …… (3 s.f.)

630

c 6103 (2 s.f.)

6100

d 0.005072 (2 s.f.)

0.0031

e 0.03926 (3 s.f.)

0.0393

f 0.56201 (2 s.f.)

0.56

Key Ideas

1. For numbers based on measurements. Not all digits have meaning, due to the limits of ……
2. Only digits with meaning are written. These digits are called …….. figures.

1 Decide how many significant figures are in these quantities.

a 1.234 4	**b** 0.001 1
c 1.2 2	**d** 0.012 2
e 1002 4	**f** 1.023 4
g 3.07 3	**h** 0.200 3

2 Complete the following to round 28.470 58 to the given level of accuracy.

a 1 s.f.

1st significant figure: 2

Round to nearest ten

28.470 58 ≈ ..... (1 s.f.)

30

b 2 s.f.

2nd significant figure: .....

Round to nearest .....

28.470 58 ≈ ..... (2 s.f.)

28

c 3 s.f.

3rd significant figure: .....

Round to nearest .....

28.470 58 ≈ ..... (3 s.f.)

28.5

3 Complete the following to round 0.037 09 to the given level of accuracy.

a 1 s.f.

1st significant figure: .....

Round to nearest .....

0.037 09 ≈ ..... (1 s.f.)

0.04

b 2 s.f.

2nd significant figure: .....

Round to nearest .....

0.037 09 ≈ ..... (2 s.f.)

0.037

c 3 s.f.

3rd significant figure: .....

Round to nearest .....

0.037 09 ≈ ..... (3 s.f.)

0.0371

4 Anna rounds 7382 to 1 significant figure. She says:



The answer is 7000. It was the same as rounding to the nearest 1000.

Explain why rounding to 1 significant figure is not always the same as rounding to the nearest 1000. Give an example.

First significant figure is not always thousands.

For example: 81, the first significant figure is 8, which is tens.

Round each of these numbers to 1 significant figure.

a 47 381 50 000	**b** 4738.1 5000	**c** 473.81 500
d 47.381 50	**e** 4.7381 5	**f** 0.473 81 0.5

5 Round the following:

a 4 170 162

Sig Figs	Rounded
1	4 000 000
2	4 200 000
3	4 170 000
4	4 170 000

b 0.003 661 2

Sig Figs	Rounded
1	0.004
2	0.003 7
3	0.003 66
4	0.003 661

c 24.8706

Sig Figs	Rounded
1	20
2	25
3	24.9
4	24.87

6 Write these numbers correct to the number of significant figures indicated.

a	1 561 231 (2 s.f.) 1 600 000	**b**	3677 720 (4 s.f.) 3 678 000	**c**	789 001 (5 s.f.) 789 000
d	3 300 000 (1 s.f.) 3 000 000	**e**	777 777 (3 s.f.) 778 000	**f**	3194 729 (5 s.f.) 3 194 700
g	0.0035 (1 s.f.) 0.004	**h**	0.191 785 (4 s.f.) 0.1918	**i**	0.001 592 (3 s.f.) 0.001 59
j	0.111 222 (5 s.f.) 0.111 22	**k**	0.000 0271 (1 s.f.) 0.000 03	**l**	0.019 832 (5 s.f.) 0.019 832

7 Most questions will ask you to round to either 2 decimal places or 3 significant figures.
For each of these numbers, round to 2 decimal places and to 3 significant figures.

	2 d.p.	3 s.f.
a 17.962	17.96	18.0
b 1.4638	1.46	1.46
c 47.859	47.86	47.9
d 804.5272	804.53	805
e 500.5749	500.57	501
f 821.2749	821.27	821
g 5810.2539	5810.25	5810
h 0.02389	0.02	0.0239
i 0.01537	0.02	0.0154

8 Round to 2 sf.

a 32

32

b 320

320

c 328

330

d 368

370

e 36.8

37

f 3.68

3.7

g 0.368

0.37

h 0.0368

0.037

i 0.000 003 68

0.000 003 7

j 0.000 013 68

0.000 014

k 4.000 013 68

4.0

l 4.000 093 68

4.0

m 4.900 093 68

4.9

n 490 009 368

490 000 000

o 4 900 093 680

4 900 000 000

p 4 090 009 368

4 100 000 000

9 Work out each of these on your calculator. Round each answer to 2 significant figures.

a 1.89×319.175

600

b $\sqrt{3} \times \sqrt{2}$

2.4

c 195×4.3

840

d $\frac{2355 \times 978}{1.95}$

1 200 000

10 The number 5□3 is rounded to 1 significant figure. The answer is 600.
What could the original number have been?

Any digit 5 to 9

11 A number rounded to 1 significant figure is 100 000.
The same number rounded to 2 significant figures is 100 000.
The same number rounded to 3 and to 4 significant figures is also 100 000.
What could the number be?

99999 (many possible answers)

12 By rounding each number to 1 significant figure, estimate the answers to these calculations.

a 58×72

4200

b 7.3×9.6

70

c 88^2

8100

d $512 \div 4.78$

100

13 The speed of light in a vacuum is 299 792 458 m/s. Round this number to:

a 1 s.f.

300 000 000

b 2 s.f.

300 000 000

c 3 s.f.

300 000 000

d What do you notice?

They're the same

14 2016 HSC Standard 2 Band 4

What is 208.345 correct to two significant figures?

A. 208

B. 210

C. 208.34

D. 208.35

B

15 2022 HSC Standard 1

The area (A) of a circle is given by the formula $A = \pi r^2$, where r is the radius.

What is the value of A, correct to three significant figures, if $r = 3.55$?

A. 39.5

B. 39.6

C. 39.591

D. 39.592

B

16 2014 HSC Advanced Band 3

What is the value of $\frac{\pi^2}{6}$ correct to 3 significant figures?

A. 1.64

B. 1.65

C. 1.644

D. 1.645

A

Scientific Notation (Standard Form)

Review

1 Complete the following:

	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
Fraction	$\frac{1}{10}$				
Decimal					

2 Complete the table. You can use your calculator to help you.

Power of 10	10^5	10^4	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
Ordinary number	100 000	10 000			10	1	0.1	0.01			

a Which powers of 10 are large?

b Which powers of 10 are small?

c Which powers of 10 are positive?

d Which powers of 10 are negative?

3

a Write one million as a power of 10.

b Write one millionth as a power of 10.

c What is the same about your answers? What is different?

ⓘ Scientific Notation (Standard Form)

- **Scientific notation**, or **standard form**, shows very large or very small numbers.
- Scientific notation expresses numbers as a multiple of a **power of 10**.

$$a \times 10^n$$

- a: Number between 1 and 10
- n: Any integer. Positive n: large number. Negative n: small number

- Identify the largest place column to have value (i.e. the first significant figure), then treat that as the new counting unit.
  - 6 500 000 "6.5 million" 6.5×10^6
  - 300 000 "3 hundred thousands" 3×10^5
  - 0.004 35 "4.35 thousandths" 4.35×10^{-3}

🔍 **Identify** numbers written in scientific notation

✗	✓	✓	✗
30×10^5	3×10^5	3.85×10^{-12}	0.85×10^{-12}

Determine whether these numbers are written in scientific notation

a	3×10^5	**b**	3×10^6	**c**	3×10^{67}	**d**	$3 \times 10^{6.7}$
e	$3 \times 10^{0.67}$	**f**	3×10^7	**g**	3×10^{-7}	**h**	$3 \times 10^{-0.7}$
i	3×100^7	**j**	30×10	**k**	30×10^7	**l**	30×10^{-7}
m	0.3×10^7	**n**	3.3×10^7	**o**	3.0×10^7	**p**	30000

💡 Key Ideas

1. Scientific notation and …… form refer to the same thing:

   $a \times 10^n$, where a is a number between …… and ……, and n is an ……

1 Decide whether these numbers have been written in scientific notation or not.

a	5.9×10^6 yes	**b**	34×10^8 no	**c**	$8.97 \times 10\,000$ no	**d**	5.03×10^{-9} yes
e	28000 no	**f**	7×10^{-15} yes	**g**	0.85×10^4 no	**h**	2.006×10^{60} yes

2 Complete the table.

Scientific notation	Expanded Form	Basic Numeral
5×10^3	5×1000	$5\,000$
3×10^4	$3 \times 10\,000$	$30\,000$
2×10^5	$2 \times 100\,000$	$300\,000$
7×10^4	$7 \times 10\,000$	$70\,000$
4×10^5	$4 \times 100\,000$	$400\,000$
2×10^5	$2 \times 100\,000$	$200\,000$

3 Complete the table.

Scientific notation	Fraction	Basic Numeral
2×10^{-4}	$\frac{2}{10^4} = \frac{2}{10\,000}$	0.0002
3×10^{-2}	$\frac{3}{100}$	0.03
5×10^{-3}	$\frac{5}{1000}$	0.005
8×10^{-2}	$\frac{8}{100}$	0.08
6×10^{-4}	$\frac{6}{10\,000}$	0.0006
4×10^{-5}	$\frac{4}{100\,000}$	0.00004

Converting to Scientific Notation

Converting to Scientific Notation

1. Move the decimal point after the first non-zero digit.
2. Count digits between the new and old decimal points. This is the power of 10.
3. Write the power of 10:
   a. Positive for large numbers,
   b. Negative for small numbers.

$3000000.$

300000.0×10^{1}

30000.00×10^{2}

3000.000×10^{3}

300.0000×10^{4}

0.000032

00.00032×10^{-1}

000.0032×10^{-2}

0000.032×10^{-3}

00000.32×10^{-4}

Example Represent numbers using scientific notation

Convert to scientific notation

246000

2.46000

$= 2.46 \times 10^{5}$

0.00071

$0\ 0007.1$

$= 7.1 \times 10^{-4}$

Guided Practice

Convert these numbers to scientific notation

a	50 000	**b**	452 000	**c**	5300
	5×10^{4}		4.52×10^{5}		5.3×10^{3}
d	0.004	**e**	0.000 57	**f**	0.000 0132
	4×10^{-3}		5.7×10^{-4}		1.32×10^{-5}

1 Write these numbers in scientific notation.

a 26 000 2.6×10^4	**b** 670 6.7×10^2	**c** 219 000 2.19×10^5
d 63 000 000 6.3×10^7	**e** 7 540 000 7.54×10^6	**f** 35 3.5×10^1
g 718 000 000 7.18×10^8	**h** 90 300 9.03×10^4	**i** 66 120 000 000 6.612×10^{10}
j 1.2 million 1.2×10^6	**k** 7155 7.155×10^3	**l** 730 000 025 000 $7.300\,000\,25 \times 10^{11}$

2 Write each of these numbers in scientific notation.

a 0.12 1.2×10^{-1}	**b** 0.012 1.2×10^{-2}	**c** 0.0012 1.2×10^{-3}	**d** 0.000 12 1.2×10^{-4}
e 0.000 03 3×10^{-5}	**f** 0.000 032 3.2×10^{-5}	**g** 0.000 031 7 3.17×10^{-5}	

3 Write the following numbers in scientific notation.

a 7600 7.6×10^3	**b** 1 700 000 000 1.7×10^9	**c** 0.000 56 5.6×10^{-4}
d 0.000 067 8 6.78×10^{-5}	**e** 35 000 3.5×10^4	**f** 0.26 2.6×10^{-1}
g 590 000 5.9×10^5	**h** 6 800 000 6.8×10^6	**i** 0.000 000 812 8.12×10^{-7}
j 0.004 3 4.3×10^{-3}	**k** 310 000 000 3.1×10^8	**l** 0.092 9.2×10^{-2}

4 Express the following numbers in scientific notation.

a The number of hairs on a person's head is approximately 129 000. .....

1.29×10^5 hairs

b The distance from Earth to the Sun is 152 000 000 km. .....

1.52×10^8 km

c The diameter of a hydrogen atom is 0.000 000 002 54 cm. .....

2.54×10^{-9} cm

d The size of the influenza virus is 0.000 000 26 m. .....

2.6×10^{-7} m

Converting to Decimal Form

Converting to Decimal Form

- For $a \times 10^n$

1. Move the decimal left or right depending on the value of n.
    - If n is positive, move the decimal n places to the right.
    - If n is negative, move the decimal n places to the left.

- **Remember**: big numbers have positive power, small numbers have negative power.

Example Represent numbers as decimal form

Convert to decimal form.

6.48×10^5

$= 6\ 48000.$

$= 648\ 000$

3.51×10^{-6}

$= 0.000003\ 51$

$= 0.000\ 003\ 51$

Guided Practice

Convert these numbers to decimal form.

a 3×10^4 30 000	**b** 5×10^{-2} 0.05	**c** 3.9×10^6 3 900 000
d 8.737×10^2 873.7	**e** 3.51×10^{-6} 0.000 003 51	**f** 2.71×10^{-5} 0.000 027 1

1 Write these numbers in decimal form.

a 2×10^4 20 000	**b** 1.5×10^3 1500	**c** 3.82×10^5 382 000
d 1.79×10^4 17 900	**e** 9.91×10^6 9 910 000	**f** 6.5×10^2 650
g 2.274×10^5 227 400	**h** 6.5823×10^8 658 230 000	**i** 8×10^5 800 000

2 Write these numbers in decimal form.

a 2.4×10^{-3} 0.0024	**b** 1.56×10^{-4} 0.000 156	**c** 8.1×10^{-2} 0.081
d 7.685×10^{-6} 0.000 007 685	**e** 1.35×10^{-7} 0.000 000 135	**f** 7.6×10^{-1} 0.76
g 5.564×10^{-4} 0.000 556 4	**h** 2.367×10^{-6} 0.000 002 367	**i** 6×10^{-3} 0.006

3 Match the cards that are equal in value.

e 2.9×10^4	**f** 2.09×10^3	**g** 2.9×10^6	**h** 2.09×10^4
A 2 900 000	**B** 29 000	**C** 2090	**D** 20 900

a – B, b – C, c – A, d – D

4 Write the following numbers in decimal form.

a 1.12×10^5 112 000	**b** 5.34×10^8 534 000 000	**c** 5.81×10^{-3} 0.00581
d 1.63×10^{-7} 0.000 000 163	**e** 2.4×10^2 240	**f** 3.5×10^{-4} 0.00035
g 7.9×10^{-6} 0.000 007 9	**h** 5.2×10^3 5200	**i** 8.678×10^7 86 780 000

5 Jackson says: "$2.3 \times 10^4 = 230\,000$"

Explain why he is incorrect and write 2.3×10^4 in decimal form.

$2.3 \times 10^4 = 2.3 \times 10\,000 = 23\,000$

Mixed Practice

FOUNDATION

1 Write >, <, or = to compare each pair of numbers.

a $3 \times 10^5 \bigcirc 2 \times 10^5$ >

b $3 \times 10^5 \bigcirc 7 \times 10^8$ <

c $3 \times 10^5 \bigcirc 340\,000$ <

d $3 \times 10^5 \bigcirc 4 \times 10^5$ <

e $3 \times 10^5 \bigcirc 6 \times 10^2$ >

2 Compare these numbers using <, >, or =

a $4.5 \times 10^3 \bigcirc 4.2 \times 10^3$ >

b $7 \times 10^4 \bigcirc 55\,000$ >

c $8.95 \times 10^5 \bigcirc 2.8 \times 10^6$ <

d $1.03 \times 10^{-2} \bigcirc 1.5 \times 10^{-2}$ <

e $1.03 \times 10^{-2} \bigcirc 1.5 \times 10^{-3}$ >

f $1.03 \times 10^{-2} \bigcirc 1.5 \times 10^{-1}$ <

3 Arrange the following sets of numbers in ascending order.

a

$6.3 \times 10^5, \quad 0.0056 \times 10^8, \quad 672\,000,$

$0.0056 \times 10^8, 6.3 \times 10^5, 672{,}000$

b

$0.000234, \quad 2.341 \times 10^{-3}, \quad 234.01 \times 10^{-5}$

$0.000234, 234.01 \times 10^{-5}, 2.341 \times 10^{-3}$

4 Write these numbers from largest to smallest.

$2.41 \times 10^6, \quad 24.2 \times 10^5, \quad 0.239 \times 10^7, \quad 2421 \times 10^3, \quad 0.02 \times 10^8$

$2421 \times 10^3, 24.2 \times 10^5, 2.41 \times 10^6, 0.239 \times 10^7, 0.02 \times 10^8$

5 Here are four number cards:

A 1.6×10^4 B 1.6×10^5 C 1.7×10^4 D 1.7×10^6

a Which card is 10 times greater than A? .....

B

b Which card is 100 times smaller than D? .....

C

c E is a thousand times greater than A. Write E in scientific notation. .....

1.6×10^7

6 Choose the correct answer to each calculation,

a $6 \times 10^4 + 3 \times 10^5$

9×10^9 | 9×10^5 | 63 000 | 360 000

360 000

b $6 \times 10^5 - 3 \times 10^4$

3×10^1 | 3×10^5 | 630 000 | 570 000

570 000

c $6 \times 10^{-2} - 3 \times 10^{-4}$

3×10^{-6} | 3×10^2 | 0.0597 | 0.00597

0.0597

d $6 \times 10^3 + 3 \times 10^{-3}$

9×10^0 | 6000.003 | 6000.0003 | 6.3×10^3

6000.003

e $6 \times 10^3 - 1 \times 10^{-3}$

5×10^{-3} | 6000.001 | 5999.999 | 5×10^0

5999.999

7 Rewrite each number in scientific notation.

a	**i** 30×10^3 3×10^4	**ii** 30×10^4 3×10^5	**iii** 30×10^5 3×10^6	**iv** 30×10^{27} 3×10^{28}
b	**i** 80×10^7 8×10^8	**ii** 800×10^7 8×10^9	**iii** 8000×10^7 8×10^{10}	**iv** $800\,000 \times 10^7$ 8×10^{12}
c	**i** 20×10^3 2×10^4	**ii** 12×10^3 1.2×10^4	**iii** 23×10^3 2.3×10^4	**iv** 105×10^3 1.05×10^5
d	**i** 80×10^{14} 8×10^{15}	**ii** 97×10^{14} 9.7×10^{15}	**iii** 230×10^{14} 2.3×10^{16}	**iv** 10.5×10^{14} 1.05×10^{15}
e	**i** 1250×10^{-3} 1.25	**ii** 125×10^{-3} 1.25×10^{-1}	**iii** 12.5×10^{-3} 1.25×10^{-2}	**iv** 0.125×10^{-3} 1.25×10^{-4}

8 Work out the missing numbers.

a $(12 \times 10^9) \div (2 \times 10^{\square}) = 6 \times 10^3$

6

b $(6 \times 10^{\square}) \div (4 \times 10^3) = 1.5 \times 10^5$

8

c $(2.4 \times 10^2) \div (2 \times 10^{\square}) = 1.2 \times 10^{-3}$

5

d $(2.4 \times 10^{\square}) \div (2 \times 10^2) = 1.2 \times 10^{-3}$

−1

9 Using index laws, calculate the following and express your answer in scientific notation.

a $(2 \times 10^3) \times (3 \times 10^6)$

6×10^9

b $(7 \times 10^{-5}) \times (4 \times 10^{-2})$

2.8×10^{-2}

c $\frac{8 \times 10^5}{2 \times 10^{-2}}$

4×10^7

d $\frac{7 \times 10^{-7}}{7 \times 10^{-3}}$

1×10^{-4}

10 A bag of ants is 4 kg. Each ant is roughly 0.000 002 kg.



a First, express 0.000 002 kg in scientific notation:

2×10^{-6}

b Determine how many ants are in each bag.
You shouldn't need a calculator if you know your index laws!

$\frac{4}{2 \times 10^{-6}} = 2 \times 10^6$

c If an average ant weighs 2 mg and there are around 20 quadrillion (20×10^{15}) ants in the world, how much do all the ants in the world weigh?

$(20 \times 10^{15}) \times (2 \times 10^{-6}) = 40 \times 10^9 = 4 \times 10^{10}$ kg

d If an average human weighs 60 kg and there are 8 billion (8×10^9) humans, how much do all the humans in the world weigh?

$60 \times (8 \times 10^9) = 480 \times 10^9 = 4.8 \times 10^{11}$ kg

e Do all the ants in the world weigh more or all the humans?

humans

11 An average adult has about 20 000 000 000 000 red blood cells.
Each red blood cell has a mass of about 0.000 000 000 1 gram.
Work out the total mass of the red blood cells in an average adult.
Give your answer in scientific notation.

$(2 \times 10^{13}) \times (1 \times 10^{-10}) = 2 \times 10^3$ g

12 The speed of light is 3×10^8 m/s. The distance to the sun is 1.5×10^{11} m.
Calculate the time, in minutes, it takes for light to travel from the sun to Earth.

$(1.5 \times 10^{11}) \div (3 \times 10^8) = 8.330$ mins

Use a calculator for the following questions.

13 If $y = 0.5x^2$, calculate the value of y in scientific notation when $x = 9.8 \times 10^{-3}$

4.02×10^{-5}

14 Given that $V = \frac{r}{\sqrt{h}}$, find the value of r in scientific notation when $V = 5 \times 10^4$ and $h = 9 \times 10^6$

1.5×10^8

15 Find x in scientific notation, given that $x^3 = 8 \times 10^{12}$

2×10^4

16 Use the formula $E = mc^2$ to find c in scientific notation, given that $E = 4.05 \times 10^{18}$ and $m = 45$.

3×10^8

17 2009 HSC Standard 2 Band 6

The mass of a sample of microbes is 50 mg.
There are approximately 2.5×10^6 microbes in the sample.

In scientific notation, what is the approximate mass in grams of one microbe?

$0.05\text{ g} \div (2.5 \times 10^6)$

$2 \times 10^{-8}\text{ g}$

18 Find the value of each letter,

a p and q are integers.

$(p \times 10^7) + (q \times 10^2) = 50\,000\,400$ $p = 5, q = 4$

b n is a positive integer less than 10

$(n \times 10^8) - (n \times 10^5) = 399\,600\,000$ $n = 4$

c x and y are integers.

$x - y = 1$

$(y \times 10^{-2}) - (6 \times 10^{-3}) = 0.044$ $y = 5, x = 6$

Scientific Notation and Significant Figures

ⓘ Scientific Notation and Significant Figures

- For a number written in scientific notation $a \times 10^n$, every digit of a is significant.
  - 1.02×10^5 has 3 significant digits.
  - 6.0×10^{-3} has 2 significant digits.

- Rounding to a significant figure and scientific notation:
  1. Write the number in scientific notation $a \times 10^n$
  2. Round a to the required significant figure.

Example Write numbers in scientific notation and round to a significant figure

Write these numbers using scientific notation to three significant figures.

2 183 000	0.001 948 2
1. $2.183\,000$	1. $0\,001.9482$
2.183×10^6	1.9482×10^{-3}
2. 2.18×10^6	2. 1.95×10^{-3}

Guided Practice

Write these numbers using scientific notation to three significant figures.

a 487 830

1.

2.

4.88×10^5

b 0.06531

1.

2.

6.53×10^{-2}

1 State the number of significant figures given in these numbers.

a	0.272	**b**	1007	**c**	0.20	**d**	1.3
	3		4		2		2
e	1.3×10^3	**f**	4.21×10^3	**g**	2.905×10^{-2}	**h**	1.700×10^{-6}
	2		3		4		4

2 Write these numbers using scientific notation and three significant figures.

a	242 300	**b**	2 829	**c**	0.000 342 76
	2.42×10^5		2.83×10^3		3.43×10^{-4}
d	0.006 859	**e**	0.172 046	**f**	326.042
	6.86×10^{-3}		1.72×10^{-1}		3.26×10^2

3 Write using scientific notation rounding to the number of significant figures in brackets.

a	47 760 (3)	**b**	4 833 160 (4)	**c**	929.502 (1)
	4.78×10^4		4.833×10^6		9×10^2
d	0.002 81 (2)	**e**	0.000 101 (1)	**f**	37.66 (2)
	2.8×10^{-3}		1×10^{-4}		3.8×10^1

4 **2020 HSC Standard 2 Band 5**

What is 0.002073 expressed in scientific notation with two significant figures?

A. 2.07×10^{-2}

B. 2.1×10^{-2}

C. 2.07×10^{-3}

D. 2.1×10^{-3}

D

5 **2019 HSC Advanced Band 3**

What is the value of π^{10} to two significant figures.

A. 9.36×10^4

B. 9.4×10^4

C. 9.36×10^5

D. 9.4×10^5

B

6 Cheryl was asked to write 80 000 in scientific notation to 2 s.f.

She wrote 8×10^5.

a Explain why this is incorrect.

This is only 1 significant figure

b What should she have written?

8.0×10^5

7 Round the following to 3 s.f.

a 2.302×10^2

2.30×10^2

b 4.897×10^{-4}

4.90×10^{-4}

c 3.996×10^6

4.00×10^6

8 Calculate, give answer in scientific notation and round to four s.f.

a $10^{-0.4}$

3.981×10^{-1}

b $\dfrac{1}{240 - 13 \times 17}$

5.263×10^{-2}

c $20000 \times (1.01)^{25}$

2.565×10^4

d $\dfrac{11.3}{\sqrt{19.5 - 14.7}}$

5.158

e $\dfrac{3\frac{2}{3} + 5\frac{1}{4}}{4\frac{1}{2} + 6\frac{4}{5}}$

7.891×10^{-1}

f $(87.3 \times 10^4) \div (0.629 \times 10^{-8})$

1.388×10^{14}

9 The mass of a proton is 1.6726×10^{-27} kg and the mass of an electron is 9.1095×10^{-31} kg.

a Calculate, correct to 4 s.f., the ratio of the mass of the proton to the mass of an electron (i.e. how many times as heavy is a proton compared to an electron?)

1836 times as heavy

b How many protons are there in 1 kg? Answer in scientific notation to 4 s.f.

5.979×10^{26}